

1. The following are the overall purposes of structural design EXCEPT.

- A. To ensure a given structure to not fail in its design lifetime, or to has an acceptably low risk of failure.
- B. To ensure that the serviceability of the structure is not impaired by excessive deflections, vibrations, and the like.
- C. To make sure that the structure is constructed with minimum initial cost by **compromising** safety and quality.
- D. To ensure economy and harmony

ANSWER: C

2. Which of the following relation is correct?

- A. Design Strength = Ultimate strength / Partial factor of safety**
- B. Design Strength = Ultimate strength + Partial factor of safety
- C. Design Strength = Ultimate strength * Partial factor of safety
- D. Design Strength = Ultimate strength – Partial factor of safety

ANSWER: A

3. All of the following are assumptions made by Euler to define critical load except;

- A. Initially, the column is perfectly straight and elastic.
- B. The load is axial and passes through the centroid of the section.
- C. The materials of the column are homogeneous and isotropic.
- D. The shortening of the column due to axial compression is taken account

ANSWER: D

4. Which statement best describes internal and outstand elements in a compression members?

- A. Internal elements are considered to be simply supported along two edges perpendicular to the direction of compressive stress.
- B. Outstand elements are considered to be attached along one edge and free on the other edge parallel to the direction of compressive stress.
- C. Internal elements are considered to be simply supported along two edges parallel to the direction of compressive stress.
- D. B & C

ANSWER: D

5. Which one of the following parameters affect the selection of buckling curves
- A. Direction of buckling
 - B. Thickness of the material
 - C. Strength/grade of steel
 - D. All

ANSWER: D

6. What is overall flexural buckling?
- A. failure occurs by excessive deflection in plane of weaker principal axis
 - B. failure occurs by excessive deflection in plane of stronger principal axis
 - C. failure occurs by twisting of member
 - D. failure caused by seismic load
 - E. ANSWER: A

7. Which of the following is true about beam-column?
- A. net end moments are zero
 - B. member subjected to axial force and bending moment
 - C. member subjected to bending moment
 - D. member subjected to axial force only

ANSWER: B

8. Which of the following statement is true?
- A. lap joint eliminates eccentricity of applied load, butt joint results in eccentricity at connection
 - B. lap joint and butt joint eliminates eccentricity at connection
 - C. lap joint results in eccentricity of applied load, butt joint eliminates eccentricity at connection
 - D. lap joint and butt joint results in eccentricity of applied load

ANSWER: D

9. Which of the following is correct for pitch and gauge of the bolts?
- A. pitch is measured along direction of load, gauge is measured perpendicular to direction of load

- B. pitch is measured perpendicular direction of load, gauge is measured along to direction of load
- C. pitch is measured along direction of load, gauge is measured along to direction of load
- D. pitch is measured perpendicular direction of load, gauge is measured perpendicular to direction of load

ANSWER: A

10. Which one of the following columns is more slender, if the column has the same length and cross-section?

- A. Both end of the column is fixed
- B. Both end of the column if pinned
- C. One end of the column is fixed and the other end is pinned
- D. None

ANSWER: B

11. Which one of the following is true about tied and spirally reinforced columns.

- A. Spiral column has a higher initial stiffness than tied columns
- B. After yielding, both spiral and tied column failed in ductile manner
- C. After the maximum loading capacity, tied column failed in brittle failure, but spiral column shows more ductility and gives warning to the impending failure.
- D. A and C

ANSWER: C

12. The diameter of the longitudinal bars of a column should never be

- A. Less than 12 mm.
- B. Greater than 16mm
- C. Less than 10mm
- D. Greater than 20mm

ANSWER: A

13. What is the main mechanism in the development of collapse in beams and frames?

- A. By inserting a plastic hinge at a plastic load limit into a statically determinate structure.

- B. By inserting a plastic hinge at maximum load limit into a statically indeterminate structure.
- C. By forming a kinematic mechanism permitting an unbounded displacement of the system can be formed.
- D. A and C

ANSWER: D

14. The yield line analysis of a slab is:

- A. A lower bound method in which the predicted failure load of a slab for a given moment of capacity may be higher than the true value.
- B. An upper bound method in which the predicted failure load of a slab for a given moment of capacity may be higher than the true value.
- C. An upper bound method in which the predicted failure load of a slab for a given moment of capacity may be lesser than the true value.
- D. None

ANSWER: B

15. In the simplified method of analysis of flat slab, distribution of moments along the column strip and middle strip respectively as a percentage of positive moment are

- A. 75 and 25
- B. 65 and 35
- C. 55 and 45
- D. 45 and 55

ANSWER: C

16. What is the purpose of moment redistribution in beam design?

- A. To reduce congested reinforcement bars
- B. In order to solve the problem some percentage of the moment at the support is redistributed to the span region
- C. To decrease the moment over the section with higher moment and increase the moment over the a region with lower moment
- D. All

ANSWER: D

17. Why slabs are designed for bending and shear?

- A. Loading on slab is perpendicular to the horizontal surface where the span/ effective depth ratio is too high
- B. The slab tends to bend in the central area whereas near the junction of slab and column
- C. Loading on slab is parallel to the horizontal surface where the span/ effective depth ratio is too high
- D. The slab tends to bend at the edge area whereas near the junction of slab and beam
- E. A and B

ANSWER: E

18. Which one is the main reason to use helical reinforcement instead of normal links in the design of reinforced column?

- A. To protect the column against seismic load
- B. To resist high amount of load
- C. To provide the outside concrete a warning signal before the sudden failure of column
- D. A and C

ANSWER: D

19. What are the advantages of the reinforcing steel bar embedded in the concrete section?

- A. Effectively taking up all the tension without separating from the concrete, since concrete is weak in tension.
- B. The bond between steel and the surrounding concrete ensures strain compatibility.
- C. The reinforcing steel imparts ductility to a material.
- D. All

ANSWER: D

20. Identify the major criteria's which should be satisfied in the design of reinforced concrete the structure.

- A. The arrangement of space, ceiling height access and traffic flow must complement the intended use.

- B. The overall cost of the structure should not exceed the client's budget.
- C. Safely support all anticipated loadings and impairs its usefulness.
- D. A structure should be designed to require a minimum of maintenance & should be maintained in a simple fashion.
- E. All

ANSWER: E

21. Identify which expression is correct about characteristic strength according to limit state method?

- A. Is the value of the strength of material below which not more than 5 percent of the test results are expected to fail.
- B. Is the value of the strength of material in which 95% of the test result expected to pass
- C. Is the value of the strength of material in which 95% of the test result expected to fail
- D. Is the value of the strength of material below which not more than 5 percent of the test results are expected to fail.
- E. A & B

ANSWER: E

22. Which of the following factor do **NOT** affects the shear strength of reinforced concrete beams?

- A. The percentage of flexural steel in a member
- B. Effective depth of the member
- C. Type of aggregates
- D. Compressive strength of concrete
- E. None

Answer: E

23. What are the conditions for provision of doubly reinforced beam?

- A. When the applied moment exceeds the moment resisting capacity of a singly reinforced beam
- B. When the dimensions b and d of the section are restricted due to architectural, structural and/or constructional purposes.

- C. When the sections are subjected to reversal of bending moment. For example piles underground water tanks, winds braces, etc.
- D. In continuous T beam where the portion of beam over middle support has to be designed as doubly reinforced.
- E. When the beams are subjected to eccentric loading, shocks or impact loads.
- F. ALL

ANSWER: F

24. In the design of reinforced concrete structural which of the following is/are correct regarding to anchorage and Splicing of reinforcement

- A. Anchorage is provided near the extreme end (or cut-off point) of a bar subjected to tension (or compression).
- B. Anchorage is essentially provided to create a bond between steel and concrete
- C. Anchorage is essentially provided to axial force between steels
- D. Lap length of reinforcement is transfer effectively the axial force from the terminating bar to the connecting (continuing) bar with the same line of action at the junction
- E. All except C

ANSWER: E

25. For 8.5kN/m^2 and 5kN/m^2 dead and imposed load RC slab respectively calculate the design load for ultimate and serviceability limit state respectively.

- A. 13.55kN/m^2 and 13.55kN/m^2
- B. 13.55kN/m^2 and 18kN/m^2
- C. 18kN/m^2 and 18kN/m^2
- D. 18kN/m^2 and 13.55kN/m^2
- E. 10.5kN/m^2 and 7.5kN/m^2

ANSWER: D

26. Which one of the following is false?

- A. The depth of the building is lowering internal compressive pressure intensity for the same opening and height of the building.
- B. Local external pressure coefficient greater than or equal to global external pressure coefficient.

- C. External pressure Coefficient is always suction Lee ward side.
- D. External pressure Coefficient may have suction on wind ward side of wall
- E. None of the above

ANSWER: D

27. Which one of the following term is not included in seismic analysis of the structure

- A. Height
- B. Mass
- C. Area
- D. site soil condition
- E. All

ANSWER: C

28. Which one is not the assumption of fully plastic moment section?

- A. The yield stresses and the modulus of elasticity have the same value in compression as in tension
- B. The material is homogeneous and isotropic in both the elastic and plastic states
- C. The plane transverse sections remain plane and normal to the transverse axis after bending
- D. The cross section of the beam is symmetrical about an axis through its centroid parallel to plane of bending
- E. None of the above

ANSWER: C

29. According to Hillerborg states that the ratio of support moment to span moment is between 1.5 and 2.5, what is the implication of this ratio?

- A. Span moment is greater than support moment
- B. It shows elastic distribution of load with limited redistribution
- C. The maximum plasticity of the support
- D. Mechanism condition is within this stage
- E. All

ANSWER: B

30. Which types of forces are generated and consider during earthquake design

- A. Vertical shear

- B. Horizontal shear
- C. Bending moment
- D. Combination of horizontal and vertical shear

ANSWER: B

31. What is the main function of the diaphragm in a building during an earthquake

- A. Transfer Self Weight of Building to soil
- B. Transfer Live load from Terrace to Ground
- C. Transfer Lateral Load into Vertical Element
- D. None of the above

ANSWER: C

32. Which one of the following not the parameter of wind pressure on the structure

- A. Shape of the structure
- B. winter or Summer
- C. Velocity of air
- D. projected Area of the building
- E. None

ANSWER: E

33. In order to secure superstructure from earthquake, which technique is most preferred and used worldwide

- A. Base isolation
- B. reinforcement provision
- C. Energy dissipation
- D. seismic damper

ANSWER: A

34. Which one of the following statement is correct about design philosophy that applied on bridge design?

- A. Load resistance factor design method considers variability only on the effect of loads.
- B. Load resistance factor design method has only three limit state to design conventional bridge.

- C. Allowable stress design method is which design bridge based on the stress in the structure at working load is allowed to exceed the yield strength of the material.
- D. In load resistance factor design method load modifier is a factor accounting for ductility, redundancy and the operational classification
- E. Allowable stress design method is suited for design of modern structures than Load resistance factor design method.

ANSER: D

35. Selection of a bridge type involves consideration of one of the following factors.

- A. Geometric and subsurface conditions of the site
- B. Functional requirements
- C. Legal considerations
- D. Economics and ease of maintenance
- E. All are correct

ANSWER: E

36. Identify the incorrect statement about permanent load on bridge structures.

- A. Earth surcharge load is the result of approach slab load placed near the top of abutment which has vertical and horizontal effect.
- B. Dead load of structural components and nonstructural component are one type of permanent load estimated by taking the unit weight and cross section of the members.
- C. Lateral earth pressure has insignificant effect on instability of abutments compared to gravity permanent and transient loads.
- D. Dead load of earth fill is considered as permanent load while designing culvert.
- E. None of the above.

ANSWER: C

37. Which one of the following is a wrong about axle load value of vehicular live loads?

- A. Design truck have total axle load of 325kN
- B. Design tandem have total axle load of 220kN
- C. Design lane load have 9.3kN/m
- D. Design truck have axle load of 145kN for rear and front axle
- E. None of the above

ANSWER: D

38. Identify the major structural advantages of Truss Girder Bridge compared to other bridge superstructure?

- A. The primary forces are axial force (stress resultants are less).
- B. Greater overall depths are permissible.
- C. Have reduced self-weight.
- D. A and C
- E. All are correct

ANSWER: E

39. Which one of the following are wrong about bearing elements

- A. A two span girder will have fixed bearing at the center support and expansion bearings at the two abutments.
- B. A continuous girder require fixed bearing at one and expansion bearing on all other supports
- C. Fixed bearings allow translation but restrict rotation movements
- D. Expansion bearings allow both rotational and translational movements
- E. None of the above

ANSWER: C

40. The stopping sight distance of a vehicle moving with 45km/hr. on the leveled road and having a coefficient of friction as 0.4 is?

- A. 48m
- B. 49m
- C. 50m
- D. 51m

ANSWER: D

41. The design of horizontal and vertical alignments, super elevation, gradient is worst affected by _____

- A. Length of vehicle
- B. Width of vehicle
- C. Speed of vehicle
- D. Height of vehicle

ANSWER: C

42. In adequate drainage system will cause on pavement structure

- A. Waves and corrugations failure in flexible pavements.
- B. Stripping failure in flexible pavements
- C. Mud pumping failure in rigid pavements
- D. Volume change of subgrade soil.
- E. All

ANSWER: E

43. For a constant value of coefficient of lateral friction, the value of required super-elevation increases with

- A. Increase in both speed and radius of curve
- B. decrease in both speed and radius of curve
- C. Increase in speed and with decrease in radius of curve
- D. decrease in speed and with Increase in radius of curve

ANSWER: C

44. Which factor will be not considered as physical factor (environment) in highway engineering rout selection?

- A. Soil type
- B. Climate condition
- C. Topography of land
- D. Management man power

ANSWER: D

45. Which of the following is the false statement?

- A. In both sag and crest vertical curves with up or down grades (similar direction) no lowest and highest points on the curve.
- B. Land acquisition cost and vehicle operation cost will increase due to improper alignment selection
- C. Maintenance cost and accident severity will increase due to improper alignment selection
- D. The selection of minimum length of a crest vertical curve is controlled by headlight SSD and drainage control

ANSWER: D

46. transition curve is essentially provided to horizontal curve in the road construction is

- A. Generate more frictional forces for stability
- B. Allow vehicles to have increased speed while driving
- C. Negate the effect of centrifugal forces
- D. Avoid abrupt change in radius from a straight line to a finite radius curve

ANSWER: D

47. The full width of land acquired before finalizing a highway, alignment is known

- A. Width of formation
- B. Right of way
- C. Carriage way
- D. Roadway

ANSWER: B

48. Which one of the following is false statement?

- A. Critical tensile stress occurs at bottom of HMA for flexible pavement.
- B. Critical tensile stress occurs at the bottom of base course for cement treated base coarse flexible pavement.
- C. Critical tensile stress occurs at bottom of PCC slab for composite pavement.
- D. None

ANSWER: D

49. Which of the following is wrong about airfield design?

- A. The determination of subgrade strength, shrinkage and swelling, frost action and mud pumping, the vertical profile of the soil types, densities and moisture contents are necessary before design.
- B. While the magnitude and configuration of the wheel loads are the dominant factors in the design of airfield pavements, load repetition is an important secondary consideration for both rigid and flexible pavements.
- C. The volume and distribution of airplane traffic are important factors in design
- D. The strength of the subgrade soil is not a big concern during design

ANSWER: D

50. One of the following is not the necessary step in the design of gravel roads.

- A. Determine the minimum thickness necessary to avoid excessive compressive strain in the subgrade.
- B. Determine the aggregate crushing value of wearing course and the subgrade.

- C. Determine the extra thickness needed to compensate for the gravel loss under traffic during the period between gravelling operations.
- D. Determine the CBR value of the subgrade

ANSWER: B

51. Based on the basic equation for flexible-pavement design given in the 1993 AASHTO design method what is the immediate final output.

- A. Structural number (structural requirement needed to sustain the design traffic loadings)
- B. Pavement roughness, cracking, patching, and rutting
- C. The layer thickness with each pavement type
- D. Structural-Layer Coefficients

ANSWER: A

52. Which of the following is not correct about the main steps to be followed in designing a new road pavement according to Ethiopian Road Authority (ERA) manual?

- A. Estimating and forecasting the amount of traffic
- B. Estimating and forecasting the cumulative number of equivalent standard axles that will use the road over the selected design life
- C. Assessing the strength of the subgrade soil over which the road is to be built
- D. Selecting the most economical combination of pavement materials and layer thicknesses that will provide satisfactory service over the design life of the pavement
- E. Start building the pavement layers as per the design and by considering the notes at the catalogue

ANSWER: E

53. Which are the two major parameters considered in the Marshall Mix design?

- A. Workability and stability
- B. Density and stability
- C. Density and durability
- D. Durability and stability

ANSWER: B

54. What is the standard load applied and temperature to the sample when the needle is immersed in the penetration test?

- A. 100 mg and 25°C
- B. 100 g and 25°C
- C. 200 mg and 20°C
- D. 200 g and 20 °C

ANSWER: B

55. Aggregates are granular mineral particles used either in combination with various types of cementing material to form concrete or alone as road base. For use of this material as a base course, the sources of aggregates for construction may not include:

- A. Natural sand and gravel deposits
- B. Crushed rock
- C. Steel slag and mine refuse
- D. Artificial and processed material
- E. Pulverized/ crushed concrete and reclaimed asphalt pavement (RAP)
- F. All are sources of base course materials.

ANSWER: F

56. What are the specific gravity values (bulk and apparent) and the water absorption for the following fine aggregate?

Weight of SSD sand 500.0 g (D)

Weight of flask with water only 623.0 g (B)

Weight of flask with sand and water 938.2 g (C)

Weight of dry sand 495.5 g (A)

- A. Bulk specific gravity is 2.68, apparent specific gravity 2.75 and water absorption is 0.91%.
- A. Bulk specific gravity is 2.75, apparent specific gravity 2.68 and water absorption is 0.91%
- B. Bulk specific gravity is 2.68, apparent specific gravity 2.75 and water absorption is 0.81%.
- C. Bulk specific gravity is 2.75, apparent specific gravity 2.68 and water absorption is 0.81%

ANSWER: B

57. When the speed of the traffic flow becomes zero, then
- A. Traffic density attains its maximum value whereas traffic volume becomes zero
 - B. Traffic density and traffic volume both attain respective maximum values
 - C. Traffic density and traffic volume both becomes zero
 - D. Traffic density becomes zero whereas traffic volume attains its maximum value

ANSWER: A

58. What is the main drawback of Greenberg's Logarithmic Model?
- A. Density tends to infinity, speed tends to zero
 - B. Mean speed tends to zero, speed tends to infinity
 - C. Density tends to zero, speed tends to infinity
 - D. Flow tends to zero, speed tends to infinity

ANSWER: C

59. Which of the following is not a parameter of traffic stream?
- A. Density of traffic
 - B. Flow of traffic
 - C. Speed
 - D. PCU

ANSWER: D

60. Time-space diagram is a graph that describes the relationships between time of the vehicle as it progress in the highway and _____
- A. green time
 - B. location of intersections
 - C. location of vehicles in a traffic stream
 - D. density of traffic stream

ANSWER: C

61. The most widely used and documented trip distribution model is:
- A. Modal split
 - B. Gravity Model
 - C. Multiple regression
 - D. Cross Classification

E. All of the above

ANSWER: B

62. When does the Greenshields model satisfy the boundary condition?

- A. When the density is approaching zero as well as when the density is approaching the jam density k_j
- B. When the density is approaching infinity as well as when the density is approaching the jam density k_j
- C. When the density is approaching zero as well as when the density is approaching infinity
- D. When the density is approaching zero as well as when the density is approaching the jam density k_j

ANSWER: A

63. Which of the following is not an application of traffic flow theory?

- A. to determine adequate lane lengths
- B. to study the interrelationships of traffic flow elements
- C. to determine PCU
- D. to determine the average delay at intersections and freeway ramp merging areas

ANSWER: C

64. Which one of the following is not true about shear strength of soils

- A. Safety of all geotechnical structures depends on shear strength of soil
- B. Tension failure is more dominant failures than shear failure in soil mechanics
- C. Based on the concept of coulombs friction law the angle between the resultant force and normal force is equivalent to friction angle of the soil
- D. Friction angle and cohesion are the two constituents of shear strength parameters
- E. Shear failure occurs in soil mass by slippage of particles due to shear stresses

Ans: B- shear is more dominant

65. Which one is true about Mohr's Circle in soil mechanics

- A. Mohr circle is the graphical representation of plane stress on infinitesimal soil mass
- B. From Mohr circle one can obtain principal stresses and maximum shear stresses
- C. It is important to transform plane stresses on any inclined planes
- D. Mohr circle is used to predict the direction of failure in triaxial compression test

E. All

Ans: E –all are correct

66. A foundation has 2m by 3m dimension, the effect of ground water table is negligible on bearing capacity if the location of ground water table is

- A. At the base of the foundation
- B. 1m above the foundation
- C. 2.5m below the foundation
- D. 1m below the foundation

Ans: C

67. If the bearing capacity of a soil is 200 kPa from shear strength criterion and 250 kPa from settlement criterion, what will be the safe bearing capacity value of the soil?

- A. 250 kPa
- B. 200 kPa
- C. 450 kPa
- D. 225 kPa

Answer: B (minimum of the two)

68. Coefficient of earth pressure at rest is

- A. Greater than active earth pressure but less than passive earth pressure
- B. Less than active earth pressure but greater than passive earth pressure
- C. Less than both the active and passive earth pressures
- D. Greater than both the active earth pressure and passive earth pressure

Answer: A

69. The quantitative determination of the stability of slopes is necessary in all of the following Civil Engineering activities except one.

- A. The design of earth dams and embankments
- B. The analysis of stability of natural slopes
- C. Analysis of the stability of excavated slopes
- D. Analysis of deep seated failure of foundations and retaining walls.
- E. None of the above

Answer: - E. none of the above

70. A drained direct shear test was carried out on a sandy soil. Under a normal stress of 50 kPa, the test specimen failed at shear stress of 35 kPa. The angle of internal friction of the sample is _____ Degree. (Round of the nearest integer).

- A. 40°
- B. 35°
- C. 30°
- D. 25°

Answer: B

71. The most appropriate triaxial test to assess the long-term stability of an excavated clay slope is

- A. Consolidated drained test
- B. Unconsolidated undrained test
- C. Consolidated undrained test
- D. Unconfined compression test

ANSWER: A

72. In the consolidated undrained triaxial test on a saturated soil sample, the pore water pressure is zero

- A. during shearing stage only
- B. at the end of consolidation stage only
- C. both at the end of consolidation and during shearing stages
- D. under none of the above conditions

Answer: B

73. OMC-SP and MDD-SP denote the optimum moisture content and maximum dry density obtained from standard Proctor compaction test, respectively. OMC-MP and MDD-MP denote the optimum moisture content and maximum dry density obtained from the modified Proctor compaction test, respectively. Which one of the following is correct?

- A. $OMC-SP < OMC-MP$ and $MDD-SP < MDD-MP$
- B. $OMC-SP > OMC-MP$ and $MDD-SP < MDD-MP$
- C. $OMC-SP < OMC-MP$ and $MDD-SP > MDD-MP$
- D. $OMC-SP > OMC-MP$ and $MDD-SP > MDD-MP$

ANSWER: B

74. As per the unified soil classification system (USCS), the type of soil represented by 'MH' is

- A. Inorganic silts of high plasticity with liquid limit more than 50 %
- B. Inorganic silts of low plasticity with liquid limit less than 50 %
- C. Inorganic silts of high plasticity with liquid limit less than 50 %
- D. Inorganic silts of low plasticity with liquid limit more than 50 %

ANSWER: A

75. Four different soils are classified as CH, ML, SP, and SW, as per the Unified Soil Classification System. Which one of the following options correctly represents their arrangement in the decreasing order of hydraulic conductivity?

- A. SW, SP, ML, CH
- B. CH, ML, SP, SW
- C. SP, SW, CH, ML
- D. ML, SP, CH, SW

ANSWER: A

76. In fine grained soils, the primary consolidation occurs over a long time; whereas in coarse grained soils, the primary consolidation occurs quickly. The reason for this is due to

- A. Compressibility
- B. Permeability
- C. Moisture content
- D. Density

ANSWER: B

77. A flow net can be used for which of the following purpose?

- A. Determination of seepage
- B. Determination of seepage pressure
- C. Determination of hydrostatic pressure
- D. All of the above

Answer: D

78. Commencing 2m below the ground surface and using the standard penetration test (SPT) split barrel sampler driven 45cm, the following blow count data was obtained in increments of 15cm: 10, 15, 19. What is the N-value to be reported on the boring log for the depth of 3m?

- A. 19 blows/cm
- B. 25 blows/cm
- C. 34 blows/cm
- D. 44 blows/cm

Answer: C

79. The in-situ vane shear test is used to measure shear strength of

- A. Very soft and sensitive clays
- B. Stiff and fissured clays
- C. Sandy soils
- D. All of the above

Answer: A

80. Which one of the following is true about soil sampling

- A. Disturbed sample is used for atterberg limit and shearing resistance tests
- B. Undisturbed sample is used for consolidation and compaction tests
- C. Samples obtained from Split Spoon Sampler is undisturbed sample
- D. The area ratio of the sampler should be less than 10% to be undisturbed sample

Answer: D

81. The net ultimate bearing capacity of purely cohesive soil

- A. Depends on the width of the footing and is independent of the depth of the footing
- B. Depends on the width as well as the depth of the footing
- C. Depends on the depth, but is independent of the width, of the footing
- D. Is independent of both the width and the depth of the footing

Answer: D

82. Which one of the following is responsible for the development of negative skin friction of a pile?

- A. Settlement of pile more than that of surrounding soil
- B. Settlement of pile less than that of surrounding soil

- C. Downward movement of surrounding soil, irrespective of pile movement
- D. Downward movement of pile irrespective of settlement

ANSWER: B

83. Which one of the following is Not the purpose of site investigation

- A. The design parameters such as strength, compressibility, permeability and other parameters used for geotechnical design.
- B. To determine the geo materials available on the site which can be re used
- C. To determine the ground or the ground water conditions that would affect the design and construction example expansive soil, collapsible soils, high ground water level....
- D. To determine the effect of changes, how the design affect adjacent properties and the ground water.
- E. None of the above

Answer E

84. A 30 cm diameter concrete pile is driven in to homogenous consolidate clay deposited ($C_u = 40 \text{ kN/m}^2$, $\alpha = 0.7$) if the embedment length is 10 m, estimate the safe load (FS.= 2.5)

- A. 10.57 kN
- B. 115.7 kN
- C. 723 kN
- D. 632 kN

ANSWER: B

85. Which one of the following is a correct statement?

- A. The efficiency of driven group piles in sand & gravel is usually greater than unity
- B. The efficiency of bored pile groups in sand & gravel will never be greater than unity
- C. The ultimate capacity of group piles in clay is determined by assuming that the pile group acts as a block.
- D. The ultimate capacity of group pile in clay is equal to the sum of the base resistance and the shaft resistance of the block made by the piles.

E. All of the above

ANSWER: E

86. What is the most effective way to prevent foundation problems on expansive soils?

- A. Adding more soil to the foundation
- B. Building a foundation with a higher load-bearing capacity
- C. Installing a drainage system around the foundation
- D. None of the above

ANSWER: C

87. What is the primary reason behind soil liquefaction, which can lead to foundation failure in earthquake-prone regions?

- A. Increased soil density
- B. Pore water pressure buildup
- C. Soil compaction
- D. Expansion of clay particles

Answer: B

88. Which of the following is a post-construction measure to control reservoir silting?

- A. Construction of check dams
- B. Construction of under-sluices in dam
- C. Vegetation Screens
- D. Mechanical stirring of sediments

ANSWER: D

89. The main advantage in arch dam is that. It allows

- A. Minimum length
- B. Minimum depth
- C. Minimum width
- D. Maximum span

Answer: C

90. Which one of the following spillways is least suited to earthen dams?

- A. Ogee spillway
- B. Chute spillway
- C. Side channel spillway

D. Shaft spill way

Answer: A

91. What do you call a component of spillway that reduces erosion at the river bed by converting storm water runoff into sheet flow?

A. Exit channel

B. Energy dissipator

C. Outlet

D. Sluice

ANSWER: B

92. Which failure occurs when the net horizontal force above any plane in the dam or at the base of the dam exceeds the frictional resistance developed at that level?

A. Overturning

B. Crushing

C. Sliding

D. BY development of tension

ANSWER: C

93. Trash racks are built for _____

A. discharging the water freely from the turbine exit to tailrace

B. preventing the turbine from ingress of floating and other materials

C. creating artificial head to store sufficient potential energy of water

D. controlling the opening of valves

ANSWER: B

94. Which element of hydroelectric power plant prevents the penstock from water hammer phenomenon?

A. Valves and Gates

B. Draft tubes

C. Spillway

D. Surge Tank

ANSWER: D

95. Which statement about surge tank is wrong?

A. Ideal location of surge tank is at the turbine inlet

- B. A decrease in load demands cause a rise in water level in surge tank
- C. Surge tanks are totally closed to avoid entry of unwanted objects to penstock
- D. Surge tanks are installed to reduce harm effects of water hammer phenomenon

ANSWER: C

96. In Cement concrete mix design, with the increase in water-cement ratio. Which one of the following statements is TRUE?

- A. Compressive strength decrease but workability increase
- B. Compressive strength increase but workability decrease
- C. Both compressive strength and workability decrease
- D. Both compressive strength and workability increase

Answer: A

97. the purpose of providing cavity wall is

- A. to prevent dampness
- B. heat insulation
- C. sound insulation
- D. all

Answer: D

98. How many layers of concrete are needed to fill a slump cone?

- A. 5 layers by volume
- B. 3 equal layers by height
- C. 3 equal layer by volume
- D. 5 layers

ANSWER: C

99. Which of the following cement is suitable for use in massive concrete such as large dams?

- A. Ordinary Portland cement
- B. Low heat cement
- C. Rapid hardening cement
- D. Sulphate resisting cement

Answer: B

100. The basic purpose of a retarder in concrete is

- A. To increase the initial setting time of cement paste in concrete
- B. To decrease the initial setting time of cement paste in concrete

- C. To render the concrete more water
- D. To improve the workability of concrete mix

ANSWER: A

101. Which one of the following factors is considered for the orientation of buildings?

- A. The direction of the prevailing winds in the area
- B. The exposure of the walls and roof of the buildings to the rays of sun
- C. The extent up to which the sunrays penetrate with the verandah
- D. All the above

ANSWER: D

102. A type of bond in brick masonry consisting of alternate courses of header and stretchers is

- A. Flemish band
- B. English band
- C. header band
- D. stretchers band

Answer: B

103. A slump test for concrete is carried out to determine:

- A. Strength
- B. Durability
- C. Workability
- D. Water content

Answer: C

104. Weep holes are provide in retaining walls

- A. To drain off water from the backfill soil
- B. To ventilate the stone masonry
- C. To add architectural beauty
- D. To increase compaction of the earth retained

Answer: A

105. Which of the below is constructed above doors, windows?

- A. Joist
- B. Purlin

- C. Lintel
- D. Arch

Answer: C

106. 10m³ C-25 concrete with cement: sand: gravel (1:1.5:2.5) mix ratio is used, which of the following is the volume of each ingredient materials respectively? (Assume water/cement ratio = 0.43 and dry volume = 1.54m³)

- A. 44.35qtl: 4.92m³: 7.95m³
- B. 44.35qtl: 4.62m³: 7.70m³
- C. 44.35qtl: 5.01m³: 8.23m³
- D. 44.80qtl: 5.01m³: 8.23m³

ANSWER: A

107. What is the first step in project planning?

- A. Establish the objective and scope
- B. Determine the budget
- C. Select the team organizational model
- D. Determine project constraints
- E. Inspect the deliverables

ANSWER: A

108. Once the project is approved and moves into the planning stage, what happens in the next phase of the project life cycles?

- A. Agreements for risk sharing need to be concluded
- B. The total risk on the project typically reduces as activities are performed without loss
- C. Risks are identified with each major groups of activities
- D. A risk response plan can be used to identify increasing levels of detailed risk analysis

Answer: D

109. Alex is a Project Manager. He is coordinating a bidder conference to allow vendors to get clarification on the work that needs to be performed. Which phase of Project Management is in progress?

- A. Plan procurements

- B. Control procurements
- C. Conduct procurements
- D. Close procurements

ANSWER: C

110. Which statement is true about indirect cost?

- A. Directly related to performance the project
- B. Those that cannot be allocated to projects
- C. Those that can be easily controlled by managers
- D. Not directly related to the products or services of a project

Answer: D

111. What is the first step in project cost management?

- A. allocate project cost estimates to individual material resources
- B. plan how costs will be managed
- C. control project costs and monitor cost performance
- D. develop several estimates of costs for different types of resources

Answer: B

112. Which of the following is true about procurement documents?

- A. Procurement documents are used to solicit proposals from prospective sellers.
- B. Invitation for Bid and Request for Proposal are two examples of procurement documents
- C. Procurement documents should be structured to facilitate accurate and complete responses from prospective sellers
- D. all of the above

ANSWER: D

113. Contractor screening is important to ensure that candidates are capable of performing the work and the number is neither excessive, which will make the proposal evaluation process difficult, nor too few, which will minimize the competition. The four key items to be used in contractor screening are _____.

- A. Capacity, experience, capability, and interest
- B. Capital, capacity, experience, and location
- C. Experience, interest, financial stability, and skills

- D. Facilities, capital, capacity, and capability
- E. Industry, experience, risk, and cost

Answer: A

114. When invitations for bid are issued to the contractors, the bids (proposals) must be submitted in a standard format because _____.

- A. The owner (buyer) has a format that he routinely uses and wants to force the contractors to meet his requirements
- B. The Uniform Commercial Code specifies that all proposals will comply with the law and the standard format
- C. The government has specified that proposals must meet the requirements of contract law and this assures the incorporation of all items
- D. The Association of Contractors has prescribed the format to which all bidders must comply to meet a legal proposal
- E. It gives the greatest assurance that all items are covered and the proposals can be evaluated in a uniform manner

ANSWER: E

115. The project closing process is essential to overall project planning. What is considered one of the most important steps in closing the project?

- A. Insuring that plans are produced to implement the project by phases.
- B. Issuing work orders so that financial data can be collected.
- C. Transition the lease of equipment to the owner
- D. Satisfying the contractual obligations

ANSWER: D

116. A new project was initiated that involves new technology and subsequently has never been done before. What type of contract would the owner want to issue to reduce or eliminate as much risk as possible?

- A. Firm fixed price
- B. . Cost plus fix fee
- C. Cost plus incentive fee
- D. Lump sum
- E. A and D

ANSWER: E

117. A project management information system consists of

- A. A project management software package operating on appropriate computer facilities
- B. Hardware, software, documents, and procedures
- C. Automated and manual methods for gathering, recording, filtering, and dissemination of pertinent information for members of a project team
- D. All of the above

ANSWER: D